(AUBVC) ∓ 1- AUBVC

POKHARA UNIVERSITY

Level: Bachelor
Programme: BE
Course: Probability and Statistics

Semester: Fall

Year : 2014
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Represent the following data by means of histogram, frequency curve and polygon. 7

Salaries	300-310	310-320	320-330	330-350	350-370	370-400
No. of Workers	7	19	28	15	12	12

b) The first two groups have 100 items with mean 45 and variance 49. If the combined group has 250 items with mean 51 and variance 130, find the mean and standard deviation of the second group. 8

2. a) A problem of statistics is given to three students A, B and C where chance of solving it is $^1/_2$, $^3/_4$ and $^2/_3$ respectively. If they independently solve it, what is the probability that 7

   i. All can solve the problem
   ii. None can solve the problem
   iii. Problem can be solved
   iv. Exactly one can solve the problem.

b) State and prove Baye's theorem for conditional probability. 8

3. a) A random variable X has the following probability function. 7

X:	2	3	4	5	6	7
F(X):	0.1	K	0.2	2k	0.3	K

Find the value of K and calculate mean and variance.

b) Customers arrive at complaint department of a store at the rate of 3 per hour. If arrivals follow a Poisson distribution, calculate the probability that 8

   i. No customer will arrive in an hour.

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ii. Two or three customers will arrive in an hour.

iii. At least three customers will arrive in an hour.

4. a) In a normal distribution 31% of the items are under 45 and 8% are over 64. Find the mean and standard deviation of the distribution. 7

b) 400 orange are taken from a large consignment and 50 are found to be bad. Estimate the percentage bad orange in the consignment and assign 95% confidence limits. 8

5. a) A sample poll of 100 voters chosen at random from all voters in a given district indicated that 55% of them were in favor of a particular candidate. Find the 95% confidence limits for the proportion of all the voters in favour of this candidate. 7

b) The average weekly wages of a sample of 200 workers in industry A was Rs. 1150 with standard deviation of Rs. 100. The weekly wages of a sample of 300 workers in industry B was Rs. 1000 with standard deviation of Rs. 50. Can we consider the weekly wages paid by industry A is higher than those paid by industry B? (Use level of significance $\alpha = 0.05$). 8

6. a) The sales figure of an item in eight shops before and after advertisement is given as: 7

Before	70	65	48	72	80	92	98	100
After	72	70	53	75	84	95	105	104

Test whether advertisement was effective at 5% level of significance.

b) The following table gives the age X and blood pressure Y of 10 women. 8

X	56	42	36	47	49	42	60	72	63	55
Y	147	125	118	128	145	140	155	160	149	150

i. Find correlation coefficient between age and pressure.

ii. Fit a regression equation of X on Y.

7. Write short notes on: (Any two) 2×5

a) Binomial distribution and condition required for binomial distribution.

b) P-value and Critical value.

c) Central limit theorem and joint probability distribution.

d) Coefficient of determination.

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